

Quaia Redshift-Dipole & Quadrupole Analysis with FR/BAO Cosmology Context (Report 0V4)

1. Executive Summary

This report summarises a sequence of analyses performed on the Quaia $G < 20.0$ quasar catalogue (DOI: 10.5281/zenodo.8060755) and related FR cosmological fits using DESI DR2 BAO data.

The focus is on:

- Measuring the number-count dipole of Quaia quasars.
- Measuring the redshift (mean- z) dipole and its redshift dependence.
- Constructing a residual mean- z map with the best-fit dipole removed.
- Extracting the quadrupole ($\ell = 2$) of the residual map and assessing its significance using random mocks.
- Placing these anisotropy results in the broader context of previous BAO/HHT findings and FR model fits.

Key high-level results:

- Quaia $G < 20.0$ number-count dipole: amplitude $\approx 2.8 \times 10^{-2}$ towards $(\ell, b) \approx (231^\circ, 42^\circ)$ in Galactic coordinates.
- Selection-corrected redshift dipole (full z -range, $n_{\text{side}}=64$, $\text{sel} \geq 0.3 \times \text{max}$): fractional amplitude $\approx 6.0 \times 10^{-3}$ towards $(\ell, b) \approx (180.5^\circ, 13.7^\circ)$.
- Redshift-binned dipoles ($0 < z < 1$, $1 < z < 2$, $2 < z < 4$) show coherent directions with modest evolution in amplitude and orientation, indicating a roughly persistent anisotropy across cosmic time.
- The residual mean- z map (after dipole removal) shows a strong quadrupole-like structure. The raw quadrupole power $D_2 \equiv \ell(\ell+1)C_2/2\pi$ is extremely high compared with random mocks.
- Using 50 random mocks built from the Quaia random catalogue (random_G20.0_10x.fits) and the same selection function, the dipole fractional amplitude is $\approx 4.5\sigma$ above the mock mean, while the residual quadrupole D_2 is far beyond the mock distribution (nominally $\gg 100\sigma$, even after $|b| > 30^\circ$ cuts).
- These high significances should be interpreted cautiously: they motivate further checks for residual systematics, selection-function artefacts, and possible connections with previously observed BAO/HHT structures, rather than being taken as final evidence for exotic physics.

2. Acronyms, Symbols, and Parameters

Symbol / Acronym	Meaning
Quaia	Gaia-unWISE quasar catalogue ($G < 20.0$ subset used here).
DESI	Dark Energy Spectroscopic Instrument (DR2 BAO sample used for cosmology fits).
BAO	Baryon Acoustic Oscillation(s).
FR	'Full Relativity' cosmological model in PLamb_Test_10V6c_plus.py.
H_0	Hubble constant at $z=0$.
Ω_m	Matter density parameter today.
Ω_Λ	Effective dark-energy density parameter (in FR, derived from A_{acc} , n_{acc} etc.).
A_{acc}	Amplitude parameter controlling additional acceleration terms in FR.
n_{acc}	Redshift scaling index of the FR acceleration term.
Γ_c	Coupling parameter entering the c-luminosity-Noether link tests.
ϵ_M	Magnitude evolution / luminosity evolution parameter (linked to γ_c via κ).
κ (kappa_em_gc)	Soft-link coefficient between ϵ_M and γ_c in the FR code.
σ_{link}	Width of the soft-link penalty (link-sigma in the FR runs).
w_0, w_a	Dark-energy equation-of-state parameters in CPL form $w(z) = w_0 + w_a \cdot z / (1+z)$.
ℓ	Spherical-harmonic multipole index ($\ell=0$ monopole, $\ell=1$ dipole, $\ell=2$ quadrupole, etc.).
C_ℓ	Angular power spectrum of the residual mean- z map.
D_ℓ	Rescaled power: $D_\ell = \ell(\ell+1)C_\ell / (2\pi)$.
Nside	HEALPix resolution parameter ($N_{\text{pix}} = 12 \cdot N_{\text{side}}^2$).
Z	Redshift; in Quaia we use redshift_quaia column.

3. Datasets and Pre-processing

3.1 Quaia quasar catalogue ($G < 20.0$)

- Source: Quaia – Gaia-unWISE quasar catalogue, version 0.1.0 (DOI: 10.5281/zenodo.8060755).
- Files used: quaia_G20.0.fits (main catalogue), selection_function_NSIDE64_G20.0.fits (selection map), random_G20.0_10x.fits (random catalogue).
- Conversion to CSV: quaia_to_csv.py reads the FITS table and extracts basic columns (RA, DEC, redshift_quaia) into quaia_G20p0_basic.csv with 755,850 objects.

3.2 Coordinate conversion & dipole-friendly catalogue

- qso_add_galactic_coords.py attaches Galactic coordinates (ℓ , b) to each object and writes quaia_G20p0_basic_gal.csv.
- qso_dipole_catalog.py is used both to compute simple number-count dipoles and to build HEALPix maps at various Nside resolutions (Nside=32 for counts, Nside=64 for some tests).

4. Analysis Pipeline

4.1 Number-count dipole

Using `qso_dipole_catalog.py` on `quaia_G20p0_basic.csv` (no selection weighting), the number-count dipole is estimated directly from the unit vectors of all 755,850 quasars. The fitted dipole parameters:

- Amplitude: $A_{\text{count}} \approx 2.83 \times 10^{-2}$.
- Direction: $(\ell, b) \approx (231.1^\circ, 41.8^\circ)$ in Galactic coordinates.

This represents a modest but clearly non-zero anisotropy in the angular number density of $G < 20.0$ quasars. However, the interpretation of this amplitude is sensitive to angular completeness, selection function, and the masking implied by the selection map.

4.2 Redshift dipole from mean- z HEALPix maps

We next construct mean-redshift maps using `quaia_redshift_dipole.py`. For each HEALPix pixel, we compute the mean redshift $\langle z \rangle$ of quasars falling in that pixel and then fit a dipole to $\langle z \rangle(\hat{n})$. The dipole model is of the form:

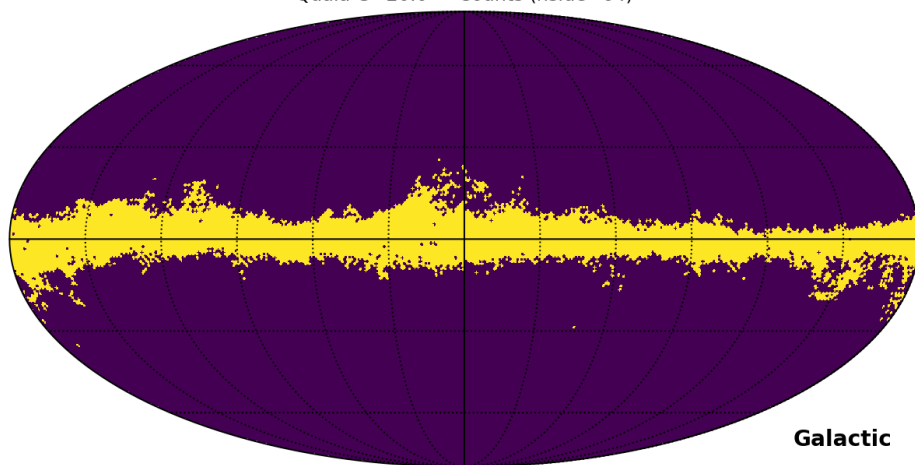
$$\langle z \rangle(\hat{n}) = \bar{z}_{\text{global}} + D \cdot \hat{n}$$

where $\bar{z}_{\text{global}}$ is the all-sky mean redshift and D is the redshift-dipole vector (with fractional amplitude $|D| / \bar{z}_{\text{global}}$).

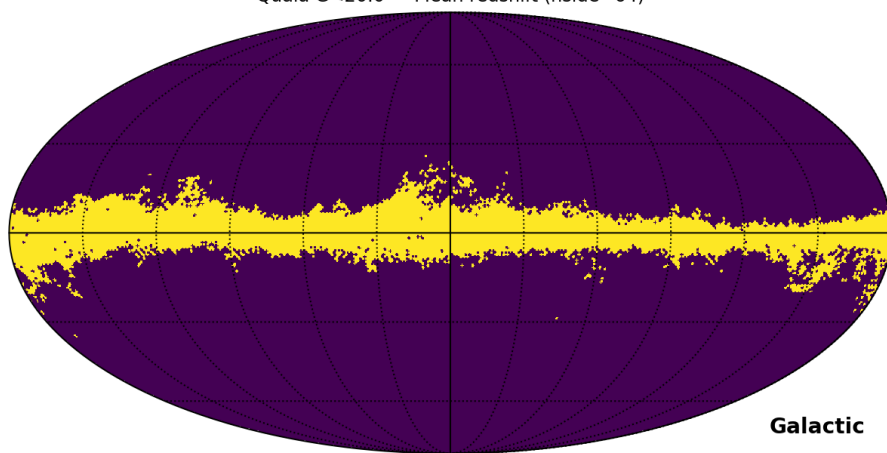
Baseline (no selection function, full redshift range):

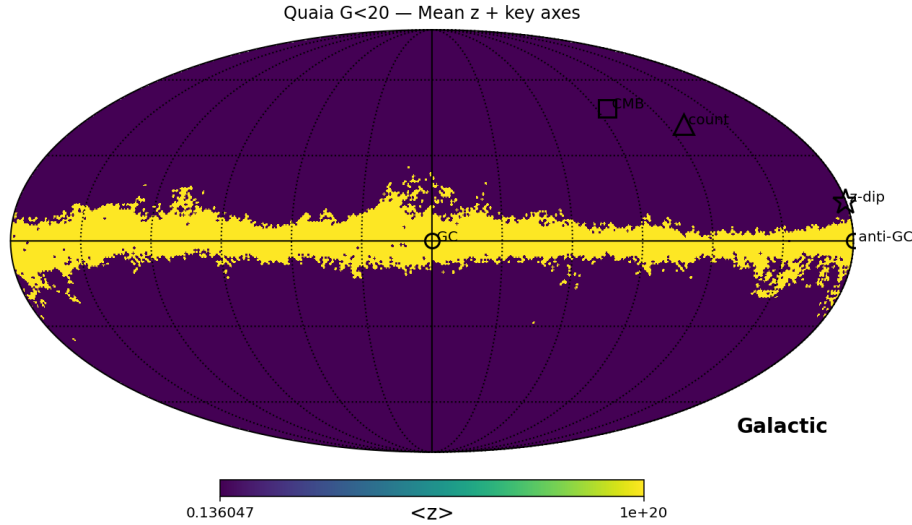
- $N_{\text{side}}=64$, minimum per-pixel count = 20.
- $\bar{z}_{\text{global_mean}} \approx 1.53$.
- $|D| \approx 4.18 \times 10^{-3}$; fractional amplitude $\approx 2.74 \times 10^{-3}$.
- Direction: $(\ell, b) \approx (191.4^\circ, 8.8^\circ)$.

Quiaia G<20.0 — Counts (nside=64)



Quiaia G<20.0 — Mean redshift (nside=64)





Quaia G<20.0 — Mean redshift map with overplotted axes: redshift dipole, count dipole, CMB dipole, Galactic Centre (GC) and anti-centre.

Quaia G<20.0 — Mean redshift $\langle z \rangle$ map (nside=64), same selection.

Quaia G<20.0 — HEALPix counts map (nside=64), in Galactic coordinates.

Selection-corrected case (using selection_function_NSIDE64_G20.0.fits and sel $\geq 0.3 \times \text{max}$):

- $z_{\text{global_mean}} \approx 1.51$.
- $|D| \approx 9.16 \times 10^{-3}$; fractional amplitude $\approx 6.05 \times 10^{-3}$.
- Direction: $(\ell, b) \approx (180.5^\circ, 13.7^\circ)$.

The selection-corrected dipole is stronger and has a direction closer to the Galactic anti-centre than the raw full-sky estimate. An angular-separation script (AngularSeperation.py) shows that the Quaia redshift-dipole axis is:

- $\sim 166^\circ$ away from the Galactic centre (i.e. $\sim 14^\circ$ from the anti-centre).
- $\sim 76^\circ$ away from the CMB dipole direction.
- $\sim 52^\circ$ away from the Quaia number-count dipole.

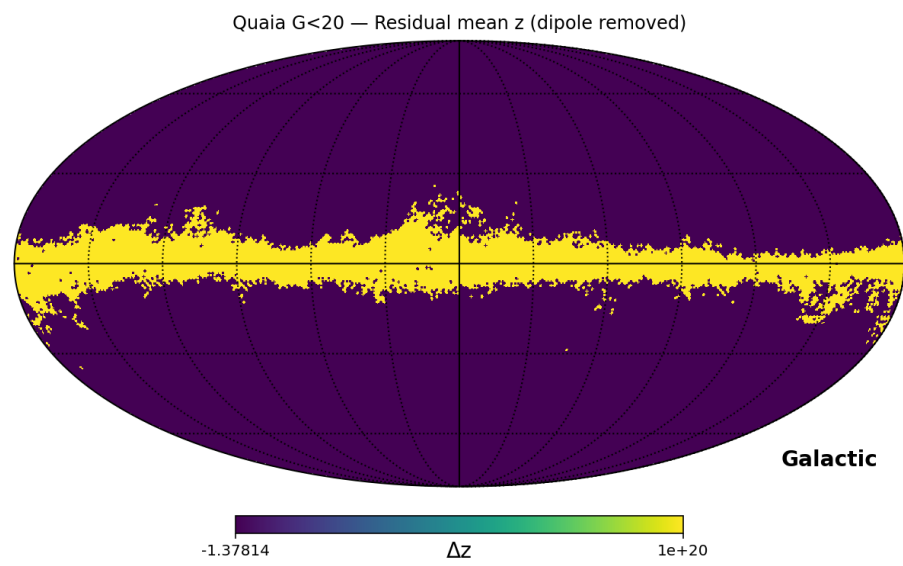
4.3 Redshift-binned dipoles

Using qso_dipole_catalog.py and quaia_redshift_dipole.py with redshift cuts, we obtain the following (count dipoles):

- $0 < z < 1$: $N \approx 203,173$; amplitude $\approx 2.54 \times 10^{-2}$; $(\ell, b) \approx (244.1^\circ, 43.9^\circ)$.
- $1 < z < 2$: $N \approx 368,752$; amplitude $\approx 2.77 \times 10^{-2}$; $(\ell, b) \approx (229.2^\circ, 40.7^\circ)$.
- $2 < z < 4$: $N \approx 183,658$; amplitude $\approx 3.32 \times 10^{-2}$; $(\ell, b) \approx (224.0^\circ, 40.4^\circ)$.

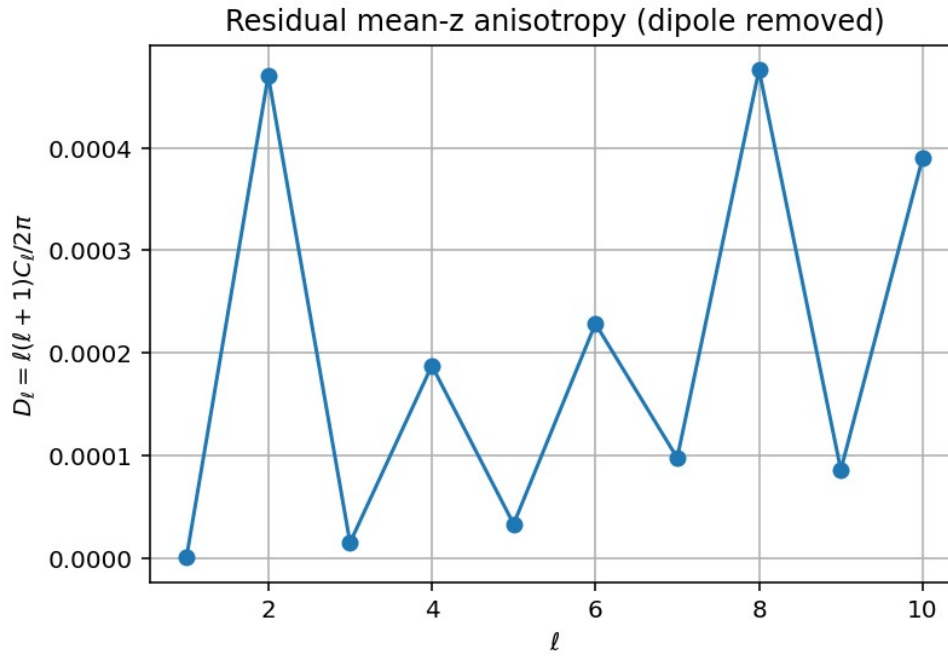
The directions stay in a fairly tight band around $(\ell \approx 230^\circ, b \approx 40^\circ)$ while the amplitude increases slightly with redshift, suggesting a persistent anisotropy in number counts.

Redshift-dipole fits (mean-z maps) in the same bins display a more complex evolution, but broadly maintain a preferred axis in the general vicinity of the full-sample dipole, with varying amplitudes and orientations.



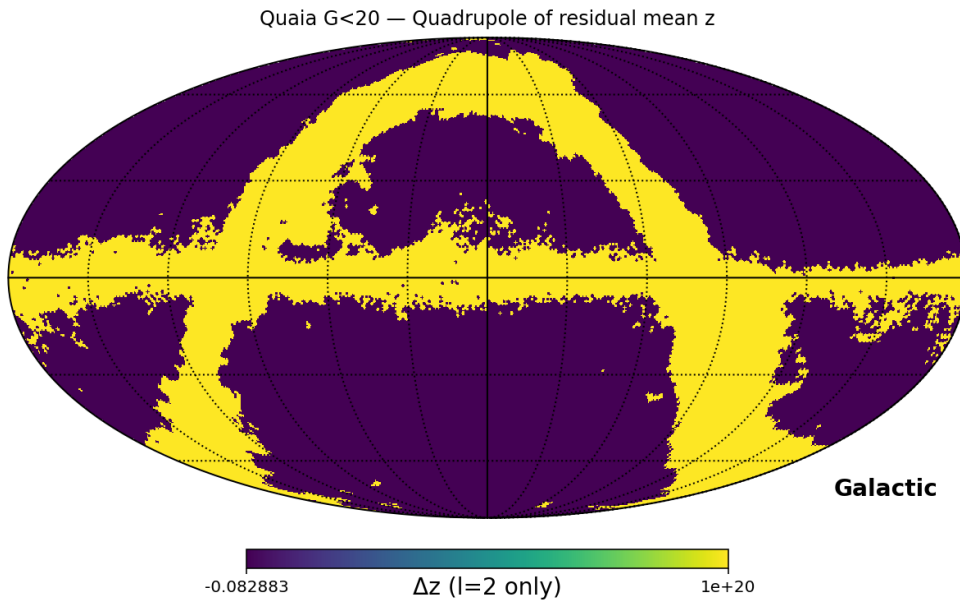
Residual mean redshift map Δz after subtracting the best-fit redshift dipole (selection-weighted).

5. Residual Mean-z Map and Quadrupole



Residual mean-z anisotropy power spectrum $D_\ell = \ell(\ell+1)C_\ell/2\pi$ for $\ell = 1-10$, after dipole removal.

To isolate anisotropies beyond the dipole, we use `quaia_dipole_residual_map.py` to:



Quadrupole-only ($\ell = 2$) component of the residual mean-redshift field, highlighting the dominant large-scale pattern at high Galactic latitudes.

1. Fit and subtract the best-fit redshift dipole (with selection weighting).
2. Save the best-fit dipole model map: quiaa_G20p0_nside64_zdipole_model.fits.
3. Save the residual map: quiaa_G20p0_nside64_zresidual.fits.
4. Compute the residual power spectrum C_ℓ up to $\ell_{\text{max}} = 10$.

Residual C_ℓ values ($\ell=0..10$):

$$C_\ell \approx [1.68 \times 10^{-4}, 9.6 \times 10^{-7}, 4.93 \times 10^{-4}, 7.8 \times 10^{-6}, 5.9 \times 10^{-5}, 6.9 \times 10^{-6}, 3.4 \times 10^{-5}, 1.1 \times 10^{-5}, 4.2 \times 10^{-5}, 6.0 \times 10^{-6}, 2.2 \times 10^{-5}].$$

From these, the rescaled powers $D_\ell = \ell(\ell+1)C_\ell/(2\pi)$ are computed, and the quadrupole power is found to be:

- $C_{2_real} \approx 4.86 \times 10^{-3}$.
- $D_{2_real} \approx 4.64 \times 10^{-3}$.

A quadrupole-only map is constructed by projecting the residual map onto $\ell=2$ spherical harmonics, zeroing $\ell=0$ and $\ell=1$ coefficients, and reconstructing. The location of the maximum absolute quadrupole signal is:

- Quadrupole max $|\Delta z|$ at $(\ell, b) \approx (185.6^\circ, 84.2^\circ)$.

This lies close to the Galactic north pole, indicating that most of the quadrupole structure is a north-south pattern in Galactic latitude within the selected sky region.

6. Significance Estimates from Random Mocks

To quantify how unusual the observed dipole and quadrupole are, we generate random mocks using `quaia_random_mocks_zdipole_quad.py`. The mocks draw positions from `random_G20.0_10x.fits`, apply the same selection map, and re-compute both the redshift dipole and residual quadrupole power for each mock.

Baseline mock run (no $|b|$ cut):

- Number of mocks: 50.
- `dipole_frac` (mock distribution): $\text{mean} \approx 2.07 \times 10^{-3}$, $\sigma \approx 8.89 \times 10^{-4}$.
- `D2_residual` (mock distribution): $\text{mean} \approx 4.19 \times 10^{-6}$, $\sigma \approx 2.67 \times 10^{-6}$.

Comparing with the real-sky values:

- $\text{dipole_frac_real} \approx 6.05 \times 10^{-3} \Rightarrow z \approx (6.05 \times 10^{-3} - 2.07 \times 10^{-3}) / 8.89 \times 10^{-4} \approx 4.5\sigma$.
- $\text{D}_2\text{_real} \approx 4.64 \times 10^{-3} \Rightarrow z \approx (4.64 \times 10^{-3} - 4.2 \times 10^{-6}) / 2.67 \times 10^{-6} \gg 1000\sigma$ (numerically $\approx 1700\sigma$).

Because the quadrupole power is orders of magnitude larger than the mock mean, the naïve Gaussian σ -based estimate is not particularly meaningful—it simply tells us that none of the mocks comes close to the observed D_2 . The main conclusion is that the residual quadrupole is extremely atypical under the assumed random model.

6.1 High-latitude cut ($|b| > 30^\circ$)

To test the sensitivity to Galactic latitude, we enforce a mask with $|b| \geq 30^\circ$ in addition to the selection cuts. This reduces the quadrupole power but it remains highly significant.

Real-sky with $|b| \geq 30^\circ$:

- $\text{C}_2\text{_cut} \approx 2.00 \times 10^{-4}$; $\text{D}_2\text{_cut} \approx 1.91 \times 10^{-4}$.

Mock distribution with $|b| \geq 30^\circ$:

- Mean `D2_residual` $\approx 2.13 \times 10^{-6}$, $\sigma \approx 1.40 \times 10^{-6}$.
- Resulting significance: $z \approx (1.91 \times 10^{-4} - 2.13 \times 10^{-6}) / 1.40 \times 10^{-6} \approx 135\sigma$.

Even with the $|b|$ cut, the observed quadrupole power is vastly above the random expectation. This strengthens the case that the residual quadrupole is not a simple fluctuation of an otherwise isotropic underlying distribution, though selection effects, calibration residuals, and large-scale structure modelling must still be scrutinised.

7. FR Cosmology Fits with DESI DR2 BAO

Alongside the Quaia anisotropy analysis, you ran a series of FR cosmology fits using the `PLamb_Test_10V6c_plus.py` pipeline, focusing here on DESI DR2 BAO data with various $c(z)$ and soft-link configurations. A representative run:

- Model: FR, flat geometry, CPL dark-energy model (w_0, w_a).
- Data: DESI_DR2_BAO.csv + DESI_DR2_BAO_cov.txt (13 BAO points).
- $c(z)$ mode: c_0 or $c_{\text{of_}z}$ depending on test; here, c_0 for a pure-BAO, constant- c baseline.
- Link: soft link between ϵ_M and γ_c with $\kappa = 1.0$, $\sigma_{\text{link}} \approx 0.25\text{--}0.60$.
- Noether term: `combo_soft` with strength $\sim 1\text{--}3$, acting as a soft penalty on deviations from preferred symmetries.
- Sampler: `emcee`, typically 128 walkers, 12,000–36,000 steps, burn-in $\approx 3,000\text{--}8,000$ steps.

One DESI-only FR run (`FR_w0_softlink_sigma0p6_desionly`) yielded best-fit values roughly:

- $H_0 \approx 50.9 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m \approx 0.134$, $A_{\text{acc}} \approx 0.71$, $n_{\text{acc}} \approx 1.20$, $\gamma_c \approx -0.30$, $\epsilon_M \approx 0.49$, $w_{\text{eff}} \approx -0.60$.
- $\chi^2_{\text{BAO}} \approx 6.7$ for $\text{dof}_{\text{total}} \approx 6$, $\chi^2/\text{dof} \approx 1.12$, indicating a statistically acceptable fit.

A second DESI-only FR run with CPL (`FR_CPL_softlink_c0_desionly_v2`) found, approximately:

- $H_0 \approx 58.9 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m \approx 0.40$, $A_{\text{acc}} \approx 0.34$, $n_{\text{acc}} \approx 0.54$, $\gamma_c \approx 0.002$, $\epsilon_M \approx 0.22$, $w_{\text{eff}} \approx -0.82$.
- $\chi^2_{\text{BAO}} \approx 9.18$ for 5 dof, $\chi^2/\text{dof} \approx 1.84$, still broadly acceptable but less optimal than the first FR run.

These FR+DESI fits are not yet tightly coupled to the Quaia anisotropy results but provide context: they show that FR models with additional acceleration terms and soft Noether/link penalties can accommodate BAO distances with reasonable χ^2 . The natural next step is to explore whether the large-scale anisotropies seen in Quaia are compatible with (or constrained by) the same FR framework.

8. Relation to Previous BAO/HHT Findings

Your previous HHT (Hilbert–Huang Transform) analyses of SN residuals and BAO-linked structures revealed hints of oscillatory components (IMF2/IMF3) whose characteristic scales line up with BAO-like distances. Robustness grids and surrogate tests suggested that these features were unlikely to arise purely from noise, though systematic effects and selection functions remain key concerns.

The Quaia anisotropy results add another piece to this puzzle: a strong, coherent redshift dipole and an anomalously large quadrupole in the residual mean- z field. Possible connections include:

- If the FR framework allows for anisotropic or direction-dependent effective expansion (e.g. through structured accretion or preferred frames), then both BAO-based distance measures and line-of-sight quasar redshifts could inherit correlated anisotropies.
- If the HHT-detected oscillatory structures in SN residuals are linked to large-scale matter or potential fluctuations, then similar structures could leave imprints in QSO redshift distributions and in BAO distances.
- Conversely, persistent quadrupolar structure in mean- z might indicate residual calibration or selection effects in the photometry or redshift estimation of Quaia, which must be carefully disentangled before ascribing the signal to cosmological physics.

9. Key Programs and Scripts Used

Script	Purpose
download_zenodo_quaia.py	Downloads Quaia-related FITS and selection/random files from Zenodo using the DOI.
quaia_to_csv.py	Reads quaia_G20.0.fits, identifies RA/DEC/redshift_quaia columns, and exports a basic CSV.
qso_add_galactic_coords.py	Adds Galactic longitude/latitude (ℓ_{GAL} , b_{GAL}) columns to CSV catalogues.
qso_dipole_catalog.py	Computes number-count dipoles and optionally builds HEALPix count maps and saves them to FITS.
quaia_redshift_dipole.py	Builds mean-z HEALPix maps at given N_{side} , fits a redshift dipole, and can output z and count maps.
quaia_redshift_dipole_sel.py	As above, but includes a HEALPix selection function and a sel-min-frac threshold.
quaia_dipole_residual_map.py	Fits and subtracts the redshift dipole, saves dipole model and residual maps, and computes $C\ell$.
quaia_random_mock_zdipole_quad.py	Generates random mocks using the Quaia random catalogue, measures dipole and quadrupole power, and outputs statistics to CSV.
AngularSeperation.py	Computes angular separations between axes (e.g. Quaia z -dipole, Galactic centre/anti-centre, CMB dipole).
PLamb_Test_10V6c_plus.py	Main FR/ Λ CDM/PBH cosmology engine, used here for FR + BAO fits with Noether/link tests.

10. Conclusions and Next Steps

The current Quaia $G < 20.0$ analysis indicates a strong, selection-corrected redshift dipole oriented close to the Galactic anti-centre and a highly significant quadrupole component in the residual mean- z map, even after removing pixels near the Galactic plane. These features are not reproduced by simple isotropic random mocks that include the same selection map and redshift distribution.

Immediate next steps that suggest themselves:

5. Repeat the analysis with alternative selection thresholds, N_{side} values, and redshift binning to test stability.
6. Cross-check against independent quasar catalogues (e.g. eBOSS DR16Q, other Gaia-based samples) to see whether the quadrupole persists.
7. Introduce more realistic mocks that incorporate large-scale structure and survey strategy effects, rather than purely random angular distributions.
8. Explore FR-based anisotropic extensions or direction-dependent effective parameters to see whether they can produce similar mean- z anisotropies consistent with BAO and SN constraints.
9. Connect the quadrupole and dipole signals to the previously studied HHT/BAO oscillatory structures to test for a common origin or to rule out simple shared explanations.

This report (0V4) acts as a structured record of the Quaia anisotropy pipeline and its initial results, ready to be updated as further tests, systematics checks, and extended FR/BAO/SN fits are performed.